

Global Diabetes and Kidney Disease Burden Intelligence Report

An Interactive Statistical Computing Dashboard

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Registration Number: SDS6/48451/2025

Course Title: Statistical Computing
Course Code: SDS6103
Program: MSc Data Science
Faculty: Faculty of Science and Technology
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22 July 2026

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Abstract

Non-communicable diseases (NCDs) remain among the leading causes of disability and premature mortality worldwide. Diabetes and chronic kidney disease contribute substantially to this burden, creating significant health, social and economic challenges, yet the datasets that describe them are often large and difficult to interpret without specialised tools.

This project presents the **Global NCD Intelligence Dashboard**, an interactive web-based application built in R and Shiny to analyse global diabetes and kidney disease Disability-Adjusted Life Years (DALYs). The dashboard integrates statistical computing with interactive, filter-linked visualisations and short-term forecasting to provide insights into disease patterns across countries, continents and WHO regions from 1990 to 2019.

The application comprises ten modules – About & Data Notes, Executive Overview, Global Disease Overview, Geographic Intelligence, Country Rankings, Global Trends, Continent Analysis, WHO Regions, Forecast & Projection, and a Data Explorer – of which the first seven share a single year-range and continent filter, so a selection made once is reflected consistently across every linked view rather than each module operating independently. An ARIMA forecasting module projects burden five years forward with explicit 80% and 95% prediction intervals. Visualisations use Plotly and Leaflet, searchable tables use DT, and every module offers a downloadable extract of its current filtered view.

A specific data limitation is disclosed directly within the dashboard and this report rather than left implicit: the Global Burden of Disease study groups diabetes mellitus and chronic kidney disease into a single combined category at the source data's level of detail, and this project's data does not permit separating the two. The dashboard demonstrates how statistical computing can transform complex public health datasets into intuitive, honestly-scoped decision-support tools, enabling researchers, policymakers and healthcare planners to explore disease burden, identify geographical disparities, monitor long-term trends and anticipate near-term trajectory with quantified uncertainty.

1 Introduction

1.1 Background

Non-communicable diseases (NCDs) are among the leading causes of morbidity and mortality worldwide, accounting for a substantial proportion of global deaths each year. According to the World Health Organization (WHO), cardiovascular diseases, cancers, chronic respiratory diseases, diabetes and chronic kidney disease contribute significantly to the global burden of disease. The increasing prevalence of these conditions is associated with population growth, ageing, urbanisation, unhealthy diets, physical inactivity, tobacco use and other lifestyle-related risk factors.

Diabetes and chronic kidney disease are closely associated conditions that continue to impose considerable health, social and economic burdens across both developed and developing countries. Monitoring their progression requires efficient statistical methods capable of analysing large-scale health datasets collected over multiple years and geographical regions.

Although several international organisations provide open-access health datasets, these datasets are often distributed in spreadsheet format and contain thousands of observations that are difficult to interpret using conventional statistical reports. Static charts and tables limit users' ability to compare countries, identify spatial patterns and investigate temporal changes effectively.

Recent advances in statistical computing and interactive data visualisation have enabled the development of dashboards that combine statistical analysis with dynamic graphics. Such dashboards improve accessibility, facilitate exploratory data analysis and enhance evidence-based decision-making.

This project develops the **Global NCD Intelligence Dashboard**, an interactive dashboard built using the R Shiny framework to visualise and analyse the global burden of diabetes and kidney disease using Disability-Adjusted Life Years (DALYs). The dashboard integrates descriptive statistics, interactive charts, maps and searchable tables to provide comprehensive insights into disease burden across countries, continents and WHO regions.

1.2 Problem Statement

Global health organisations publish extensive datasets describing the burden of non-communicable diseases across different countries and years. However, these datasets are often large, complex and difficult to interpret without specialised statistical and computational skills. Traditional statistical reports generally provide static summaries that limit users' ability to explore geographical differences, compare countries and evaluate disease trends over time.

The absence of integrated interactive analytical tools reduces the effectiveness of data-driven decision-making among researchers, policymakers and healthcare planners. Consequently, there is a need for an interactive statistical computing solution capable of transforming complex disease burden datasets into accessible visual information that supports exploration, comparison and evidence-based public health planning.

A further, more specific problem motivates this project. The Global Burden of Disease (GBD) study groups diabetes mellitus and chronic kidney disease into a single Level 2 cause category, “Diabetes and kidney diseases,” with the two conditions broken out as separate Level 3 causes underneath it. Our World in Data’s public export of this data exposes only the combined Level 2 total. A dashboard built on this figure without disclosing the combination risks implying a level of diagnostic precision the underlying data does not support, and risks recommending the same intervention for two conditions that have different risk factors and different prevention strategies. This project treats that combination as a stated, disclosed limitation rather than an implicit assumption.

1.3 General Objective

To develop an interactive statistical computing dashboard for analysing and visualising the global burden of diabetes and kidney disease using Disability-Adjusted Life Years (DALYs).

1.4 Specific Objectives

1. To summarise the global burden of diabetes and kidney disease using descriptive statistical methods.
2. To visualise the geographical distribution of disease burden using interactive maps.
3. To compare disease burden across countries, continents and WHO regions, with all dashboard views linked to a shared set of year-range and continent filters.
4. To analyse temporal trends in diabetes and kidney disease burden from 1990 to 2019, and to produce a short-term forecast with an explicit, quantified prediction interval.
5. To develop an interactive dashboard that supports data exploration and evidence-based decision-making using statistical computing techniques, with a downloadable filtered extract available from every module.
6. To disclose and document the extent to which diabetes and kidney disease can, or cannot, be analysed as separate conditions within the available data.

1.5 Research Questions

Each question below is stated against a specific intended user of the dashboard, so that every dashboard module can be justified by the decision it supports rather than by the existence of the underlying data alone.

1. **For a global health surveillance analyst** – how has the global burden of diabetes and kidney disease changed between 1990 and 2019?
2. **For a resource-targeting planner at a ministry of health** – which countries carry the highest burden, and how does this ranking change when the year range or continent filter is narrowed?
3. **For a regional policy coordinator** – how do continents and WHO regions compare, and which are growing fastest relative to the global trend rather than merely largest in absolute terms?
4. **For an epidemiologist assessing data quality** – to what extent does the pre-combined diabetes-and-kidney-disease figure limit the conclusions that can be drawn, compared with the two conditions analysed separately?
5. **For a planning team preparing a resource case** – what does a 5-year ARIMA forecast suggest about the near-term trajectory of burden, and how much uncertainty attaches to that forecast?

1.6 Significance of the Study

This study demonstrates the application of statistical computing techniques in transforming complex health datasets into meaningful visual information that supports decision-making. The developed dashboard provides researchers with an efficient platform for exploratory data analysis, enables policymakers to identify geographical disparities in disease burden and assists healthcare planners in monitoring global health trends.

The project also demonstrates the practical application of R programming, statistical computing, interactive visualisation and dashboard development within the field of data science. Furthermore, it provides a reproducible framework that can be extended to analyse other non-communicable diseases using publicly available global health datasets.

1.7 Scope of the Study

This study focuses on the analysis and visualisation of the global burden of diabetes and chronic kidney disease using Disability-Adjusted Life Years (DALYs), reported as a single combined measure as supplied by the data source (see Section 3.3). The dashboard utilises publicly available data covering the period from 1990 to 2019 and provides analyses at the country,

continent and WHO regional levels, with a shared year-range and continent filter linking these views together, plus a 5-year ARIMA forecast with explicit prediction intervals. The project is primarily descriptive and comparative; it does not perform causal inference, and differences observed between countries or regions should not be read as evidence of a specific cause.

2 Literature Review

2.1 Introduction

This chapter reviews the existing literature related to non-communicable diseases (NCDs), statistical computing, interactive data visualization, dashboard technologies and global disease burden analysis. It also discusses the application of Disability-Adjusted Life Years (DALYs) in measuring disease burden and highlights the role of statistical computing in supporting evidence-based public health decision-making. The review establishes the theoretical and practical foundation for the development of the Global NCD Intelligence Dashboard.

2.2 Global Burden of Non-Communicable Diseases

Non-communicable diseases (NCDs) account for nearly three-quarters of all deaths globally and represent one of the greatest public health challenges of the twenty-first century. According to the World Health Organization (WHO), cardiovascular diseases, cancers, chronic respiratory diseases, diabetes and chronic kidney disease contribute substantially to disability and premature mortality. The increasing prevalence of these conditions is driven by ageing populations, urbanisation, unhealthy diets, tobacco use, alcohol consumption and physical inactivity.

Diabetes has become one of the fastest-growing chronic diseases worldwide, with complications including kidney disease, cardiovascular disease and blindness. Chronic kidney disease is closely associated with diabetes and hypertension and contributes significantly to the global burden of disease. Understanding the geographical distribution and temporal trends of these diseases is essential for designing effective prevention and intervention strategies.

2.3 Disability-Adjusted Life Years (DALYs)

Disability-Adjusted Life Years (DALYs) are widely used to quantify the burden of disease within populations. Developed through the Global Burden of Disease (GBD) study, DALYs combine Years of Life Lost (YLL) due to premature mortality and Years Lived with Disability (YLD) into a single measure.

Unlike mortality statistics alone, DALYs capture both fatal and non-fatal health outcomes, making them a comprehensive indicator of disease burden. Governments, researchers and international organisations use DALYs to compare diseases, evaluate healthcare interventions and allocate healthcare resources effectively. The present study utilises DALYs to assess the global burden of diabetes and chronic kidney disease across countries, continents and WHO regions.

2.4 Statistical Computing in Public Health

Statistical computing involves the application of computational methods for data management, statistical analysis, modelling and visualisation. Modern public health datasets often contain millions of observations collected over long periods and multiple geographical regions, making computational tools essential for efficient analysis.

The R programming language has become one of the leading platforms for statistical computing because of its extensive collection of packages supporting data manipulation, visualisation and reproducible research. Packages such as **dplyr**, **ggplot2**, **plotly**, **leaflet**, **DT** and **Shiny** provide powerful tools for building interactive analytical systems capable of exploring complex health datasets.

2.5 Interactive Dashboards for Health Data Analysis

Interactive dashboards have transformed the way healthcare data are analysed and communicated. Unlike traditional statistical reports, dashboards allow users to filter information, compare regions, visualise trends and interact with data in real time.

Several international organisations, including the World Health Organization (WHO), the Institute for Health Metrics and Evaluation (IHME) and Our World in Data, provide interactive dashboards for monitoring disease indicators. However, many existing dashboards focus on descriptive presentation rather than integrating multiple analytical perspectives within a single platform.

The Global NCD Intelligence Dashboard developed in this project combines executive summaries, geographic intelligence, temporal trend analysis, country rankings, continent comparisons, WHO regional analysis, short-term forecasting and interactive data exploration into one comprehensive decision-support system, with a shared filter linking the descriptive and comparative modules together rather than treating each as an isolated view.

2.6 Data Visualisation in Statistical Computing

Data visualisation is an essential component of statistical computing because it enables users to understand patterns, trends and relationships within large datasets. Graphical methods often reveal important insights that may remain hidden in numerical summaries.

The **ggplot2** package provides a flexible grammar of graphics for producing publication-quality statistical graphics, while **Plotly** extends these visualisations by adding interactivity such as zooming, hovering and filtering. **Leaflet** enables the creation of interactive geographical maps, allowing users to visualise spatial patterns of disease burden across countries.

The integration of these visualisation tools enhances exploratory data analysis and improves communication between researchers, policymakers and public health practitioners.

2.7 R Shiny Framework

Shiny is an open-source web application framework for the R programming language that enables developers to build interactive web applications without requiring extensive knowledge of HTML, CSS or JavaScript.

A typical Shiny application consists of a user interface (UI) and a server component. The user interface defines the layout and interactive controls, while the server processes user requests and generates dynamic outputs. Shiny has become widely adopted in healthcare, finance, education and business intelligence because of its simplicity, flexibility and ability to support reproducible statistical analysis.

The dashboard developed in this project was implemented using the Shiny framework together with Plotly, Leaflet and DT to provide an integrated environment for exploring diabetes and kidney disease burden worldwide.

2.8 Research Gap

Although numerous health organisations provide global disease datasets and visualisation platforms, many available systems are designed for broad disease surveillance and do not specifically integrate statistical summaries, interactive maps, country rankings, continent comparisons, WHO regional analyses and searchable datasets within a single dashboard focused on diabetes and chronic kidney disease.

Furthermore, many publicly available datasets remain difficult for non-technical users to analyse because they require advanced statistical skills. This project addresses these limitations by developing an interactive statistical computing dashboard that transforms complex disease burden data into accessible visual information for researchers, policymakers and healthcare planners.

A second, more specific gap concerns how such dashboards handle a data limitation that is common but rarely disclosed: several publicly available NCD burden datasets, including the one used here, report certain related conditions as a single combined figure rather than as separately interpretable measures. Dashboards built on this kind of data typically either ignore the combination silently or omit any forward-looking (forecasting) component altogether, leaving the user to assume a precision the data does not support, and to know only the past rather than the likely near-term trajectory. This project treats both points as design requirements rather than afterthoughts: the combined nature of the diabetes-and-kidney-disease measure is disclosed within the application itself, not only in accompanying documentation, and a forecasting module with explicit prediction intervals is included alongside the descriptive and comparative modules, so that uncertainty about the future is quantified rather than left unaddressed.

2.9 Chapter Summary

This chapter reviewed the literature on non-communicable diseases, Disability-Adjusted Life Years (DALYs), statistical computing, interactive dashboards and data visualisation. The review demonstrates the growing importance of statistical computing techniques in analysing large public health datasets and highlights the value of interactive dashboards for supporting evidence-based decision-making. The identified research gap provides the motivation for developing the Global NCD Intelligence Dashboard presented in this study.

3 Methodology

3.1 Introduction

This chapter describes the methodology adopted in the development of the Global NCD Intelligence Dashboard. It outlines the research design, data source, data collection, data pre-processing, statistical computing techniques, software and tools used, dashboard development process and methods used for evaluating the application.

3.2 Research Design

This study adopted a quantitative research design based on secondary data analysis. Statistical computing techniques were applied to clean, transform, analyse and visualise publicly available health data. The final output was an interactive web-based dashboard developed using the R programming language and the Shiny framework.

3.3 Data Source

The study utilised publicly available data obtained from the **Our World in Data (OWID)** repository, which compiles information from the Global Burden of Disease (GBD) study. The dataset contains estimates of Disability-Adjusted Life Years (DALYs) attributable to diabetes and chronic kidney disease for countries worldwide from 1990 to 2019.

The dataset includes variables such as:

- Country (Entity)
- Country Code (ISO3)
- Continent
- Year
- Diabetes and Kidney Disease DALYs (*a combined measure – see note below*)

The data were selected because they provide comprehensive and internationally comparable measures of disease burden across countries and over time.

A methodological limitation is disclosed here rather than left implicit. In the GBD cause hierarchy, “Diabetes and kidney diseases” is a combined Level 2 category; diabetes mellitus and chronic kidney disease exist as separate Level 3 causes beneath it, but OWID’s public export exposes only the Level 2 total. This dataset was checked directly and confirmed to contain no column that separates the two conditions. Consequently, every figure in this report and dashboard describing “diabetes and kidney disease burden” is a combined measure,

not a diabetes-specific or CKD-specific one. Separating them would require querying the GBD Results Tool (vizhub.healthdata.org/gbd-results) directly for “Diabetes mellitus” and “Chronic kidney disease” as distinct causes – noted here as a specific, actionable direction for future work (Section 5.6) rather than an unstated assumption.

3.4 Data Preprocessing

Several preprocessing steps were performed before analysis and visualisation. These included:

- Importing the datasets into R using the **readr** package.
- Removing OWID aggregate rows (codes prefixed `OWID_`) from the country-level dataset, so continent and country summaries are not double-counted.
- Handling missing values.
- Standardising country names and ISO country codes, and assigning each country to a continent using the **countrycode** package.
- Separating World Health Organization regional rows (entities tagged (`WHO`)) from country-level rows, since these are reported as pre-aggregated regions in the source data rather than derived by summing countries.
- Structuring the pipeline so that a single long-format country-level dataset is loaded once and filtered reactively inside the dashboard (by year range and continent), rather than reading independent, pre-aggregated per-module files. This is what allows a filter change on one tab to be reflected consistently across Overview, Disease Overview, Map, Rankings, Trends and Continent tabs.
- Preparing a full 1990-2019 time series (independent of the dashboard’s year-range filter) for the ARIMA forecasting step, since a shortened series would understate the model’s uncertainty.

The processed datasets were exported as CSV files for use in static reporting (this document), while the dashboard itself reads the underlying long-format data directly so that filtering stays live.

3.5 Statistical Computing Techniques

The project applied descriptive statistical methods to summarise and explore the disease burden. These techniques included:

- Frequency summaries
- Totals

- Means
- Country rankings
- Temporal trend analysis
- Continental comparisons
- WHO regional comparisons
- Short-term time series forecasting using an ARIMA (Autoregressive Integrated Moving Average) model, fitted with the `auto.arima()` procedure from the **forecast** package, producing a 5-year projection with 80% and 95% prediction intervals

These methods enabled efficient exploration of the distribution and evolution of diabetes and kidney disease burden across different geographical regions, and a statistically grounded, uncertainty-quantified view of its likely near-term trajectory.

3.6 Data Visualisation Techniques

Several visualisation methods were used to communicate the results effectively.

The dashboard includes:

- Interactive line charts for temporal trends.
- Interactive bar charts for country and disease comparisons.
- Choropleth maps displaying geographical distribution of disease burden.
- Executive summary value boxes.
- Interactive searchable tables.
- Country ranking visualisations.

The visualisations allow users to zoom, filter, search and interact with the data.

3.7 Software and Packages

The dashboard was developed entirely using the R programming language. The principal software packages used in the development of the dashboard are summarised below.

Package	Purpose
shiny	Interactive web application framework
shinydashboard	Dashboard layout and interface
dplyr	Data manipulation
readr	Reading CSV datasets
ggplot2	Statistical graphics
plotly	Interactive visualisations
leaflet	Interactive geographic maps
DT	Interactive data tables
scales	Formatting numerical values
sf	Spatial data processing
rnatrualearth	World boundary data
countrycode	Mapping ISO country codes to continents
forecast	ARIMA time series forecasting
tidyr	Reshaping data for growth/forecast tables

3.8 Dashboard Development

The dashboard was organised into ten interactive modules:

1. About & Data Notes
2. Executive Overview
3. Global Disease Overview
4. Geographic Intelligence
5. Country Rankings
6. Global Trends

7. Continent Analysis
8. WHO Regions
9. Forecast & Projection
10. Data Explorer

Each module provides a different analytical perspective. Unlike the original version of the dashboard, in which most modules were self-contained, seven of these modules (Overview, Disease Overview, Map, Rankings, Trends, Continent, and implicitly the Data Explorer's country-level view) now share a single pair of global controls – a year-range slider and a continent multi-select – so that narrowing the range in one module is reflected in every other linked module. WHO Regions is intentionally excluded from the continent filter, since WHO regions are themselves pre-aggregated groupings rather than derived from individual countries. The Forecast module uses its own scope control (global or a single selected country) and always trains on the full 1990-2019 series regardless of the shared year-range filter, since a shortened training window would understate the model's genuine uncertainty. Every module now includes a download button for its current, filtered view.

3.9 Dashboard Architecture

The application follows the standard Shiny architecture consisting of two major components:

- User Interface (UI), which defines the dashboard layout, menus, tabs and interactive controls.
- Server, which performs statistical computations, generates visualisations and updates outputs according to user interaction.

This separation improves maintainability, scalability and readability of the application.

3.10 Evaluation of the Dashboard

The developed dashboard was evaluated through functional testing. Each component was tested to verify that:

- All datasets loaded successfully.
- Interactive charts rendered correctly.
- Maps displayed geographical information accurately.
- Search and filtering functions operated correctly.
- Download functionality worked as expected.

- Navigation between dashboard tabs was successful.

The evaluation confirmed that the dashboard met the intended functional requirements.

3.11 Chapter Summary

This chapter described the methodology used in developing the Global NCD Intelligence Dashboard. It presented the research design, data source, preprocessing procedures, statistical computing techniques, visualisation methods, software tools and dashboard architecture. The next chapter presents the results obtained from the developed dashboard and discusses the key findings.

4 Results and Discussion

4.1 Introduction

This chapter presents the results obtained from the Global NCD Intelligence Dashboard. The dashboard was developed to analyse and visualise the global burden of diabetes and chronic kidney disease using Disability-Adjusted Life Years (DALYs). The results are presented according to the major dashboard modules and are discussed with respect to their usefulness in supporting public health decision-making. Unless stated otherwise, “filtered” figures below reflect the shared year-range and continent controls described in Section 3.8; changing either control updates every linked module simultaneously.

4.2 About & Data Notes

This module states, in the dashboard itself rather than only in this report, who the dashboard is for and what its central data limitation is. It names three intended uses – surveillance (tracking how burden has changed), resource targeting (identifying where burden is concentrated), and planning (using the forecast to anticipate near-term trajectory) – and discloses that diabetes and kidney disease are combined at source (Section 3.3), stating explicitly what this does and does not allow a user to conclude.

Figure 4.1: About & Data Notes

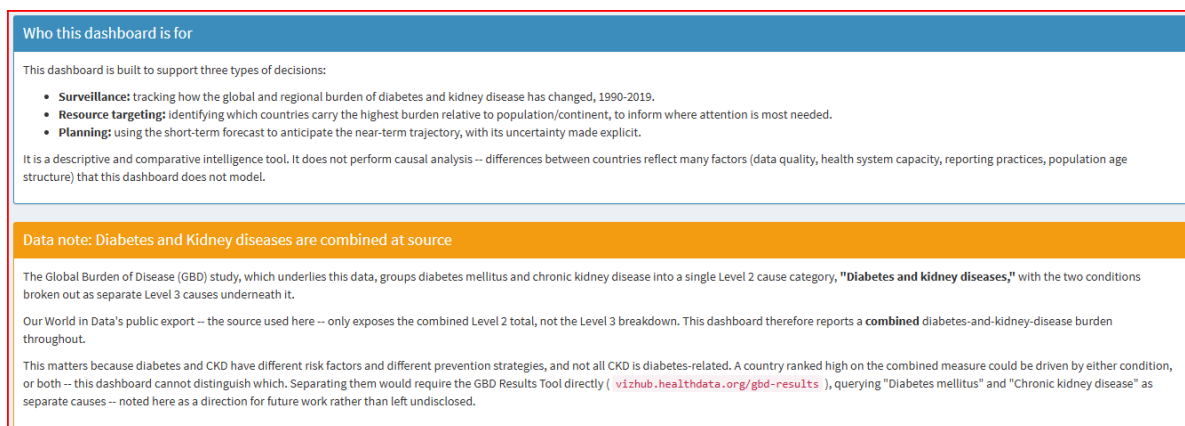


Figure 1: About & Data Notes module showing the purpose of the dashboard and important information regarding the dataset used.

4.2.1 Discussion

Placing this disclosure inside the application, not only in the appendix of a report a user may never open, means the limitation is visible at the point of use rather than discoverable only by a marker or a careful reader.

4.3 Executive Overview

The Executive Overview provides a high-level summary of the filtered dataset through interactive value boxes, trend visualisations, disease burden summaries and a ranking table. It presents the number of countries and years in the current filter selection, the total filtered DALYs and the maximum recorded DALYs. These indicators provide users with an immediate understanding of the size and scope of whatever slice of the data they have selected, rather than of the dataset as a whole.

The global trend plot illustrates changes in diabetes and chronic kidney disease burden over the filtered period, while the disease distribution chart summarises the relative contribution of the disease categories included in the analysis, recalculated for the current filter. The Top 10 Countries table identifies countries with the highest disease burden during the most recent year within the selected range, and can be downloaded directly from this tab.

Figure 4.2: Executive Overview Dashboard

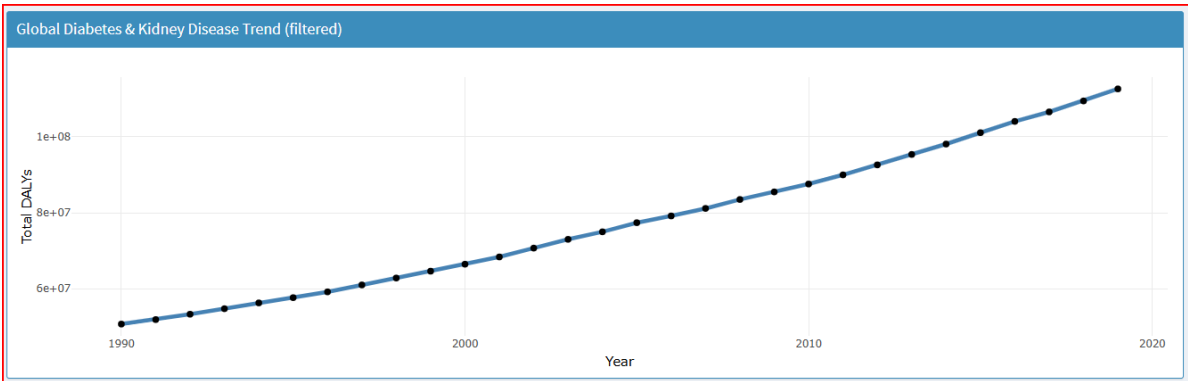


Figure 2: (a): Executive Overview – Summary Indicators and Global Trend

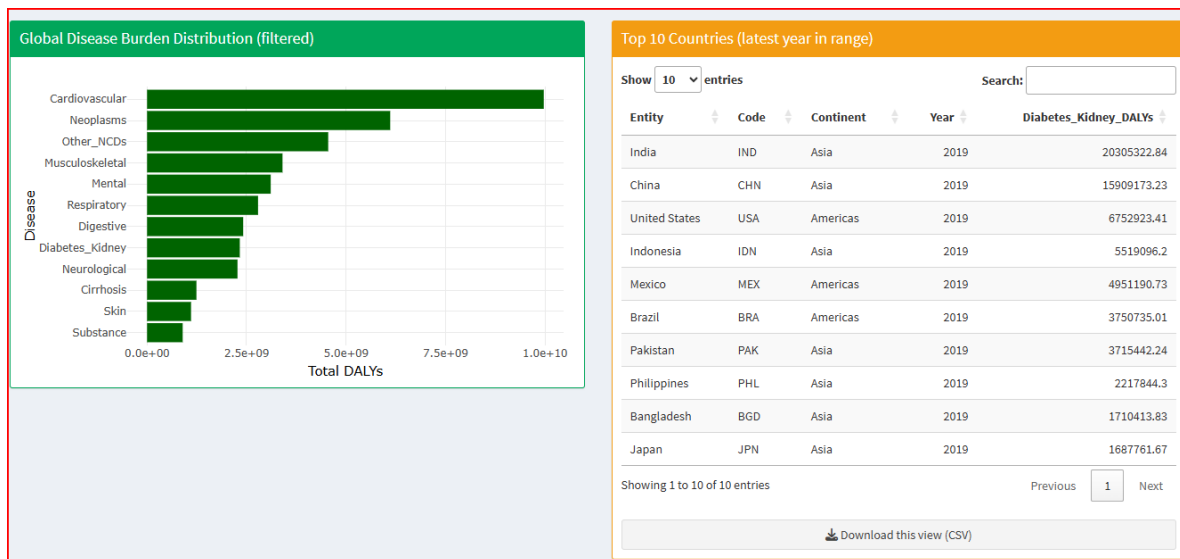


Figure 3: (b): Executive Overview – Disease Burden Distribution and Country Rankings

4.3.1 Discussion

The Executive Overview demonstrates that interactive summary statistics allow users to obtain important information quickly without manually exploring the dataset. Making the value boxes and charts respond to the shared filters – rather than always reporting the full 1990-2019, all-country picture – means this module now answers “what does the burden look like for the slice I care about,” which is closer to what a real user of the dashboard would actually ask.

4.4 Global Disease Overview

The Global Disease Overview provides descriptive statistics and graphical summaries of disease burden across all eleven NCD categories in the source data, recalculated for the current year-range and continent filter. Interactive bar charts allow users to compare diabetes and kidney disease burden against other major NCD categories (cardiovascular disease, cancers, mental disorders, and others), while searchable tables enable detailed exploration of the underlying data.

Figure 4.3: Global Disease Overview (filtered)

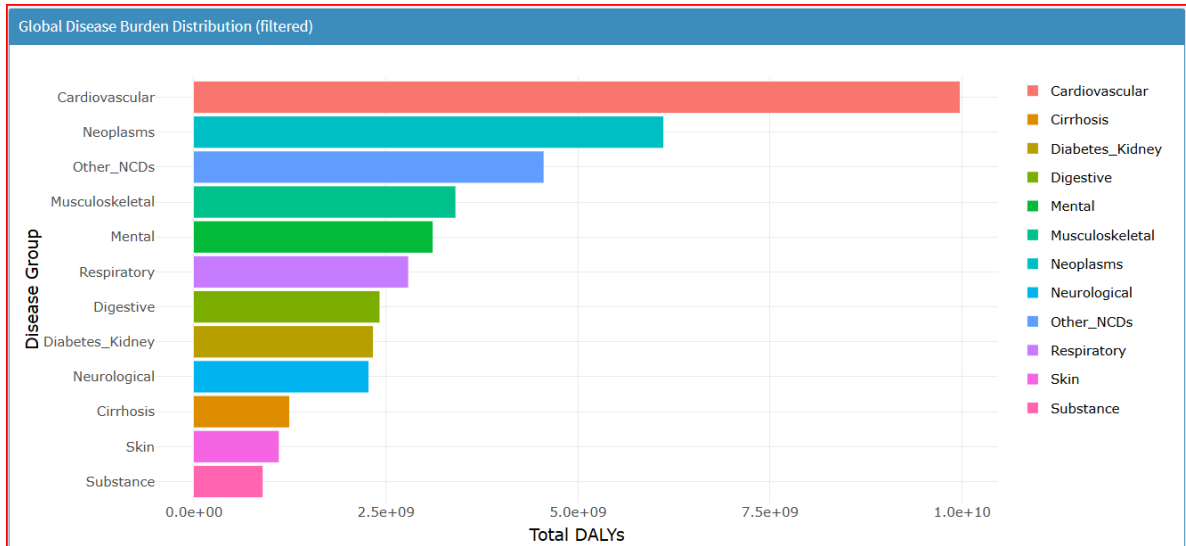


Figure 4: (a): Global Disease Overview – Distribution of Disease Burden by Major Non-Communicable Diseases

Disease Burden Dataset (filtered)

Show12entries

Search:

Disease	Total_DALYs	Percentage
Cardiovascular	9970959077.74	24.8
Neoplasms	6111551074.8	15.2
Other_NCDs	4552992209.54	11.33
Musculoskeletal	3404364675.63	8.47
Mental	3108005329.03	7.73
Respiratory	2789697109.57	6.94
Digestive	2416313379.15	6.01
Diabetes_Kidney	2330796213.32	5.8
Neurological	2273055841.14	5.65
Cirrhosis	1241207207.16	3.09
Skin	1105329002.92	2.75
Substance	895355524.35	2.23

Showing 1 to 12 of 12 entries

Previous1Next

Download this view (CSV)

Figure 5: (b): Global Disease Overview – Disease Burden Dataset Summary

4.4.1 Discussion

Interactive charts improve understanding of disease distribution by enabling users to compare disease burden visually. Searchable tables further enhance data exploration by allowing users to locate specific observations efficiently.

4.5 Geographic Intelligence

The Geographic Intelligence module presents an interactive world map illustrating the spatial distribution of diabetes and chronic kidney disease DALYs across countries, for the latest year within the selected year range and restricted to the continents selected in the shared filter. Colour gradients represent variations in disease burden, while interactive pop-up windows provide detailed country-level information. The current filtered view can be downloaded directly from this tab.

Figure 4.4: Geographic Intelligence Dashboard (filtered)

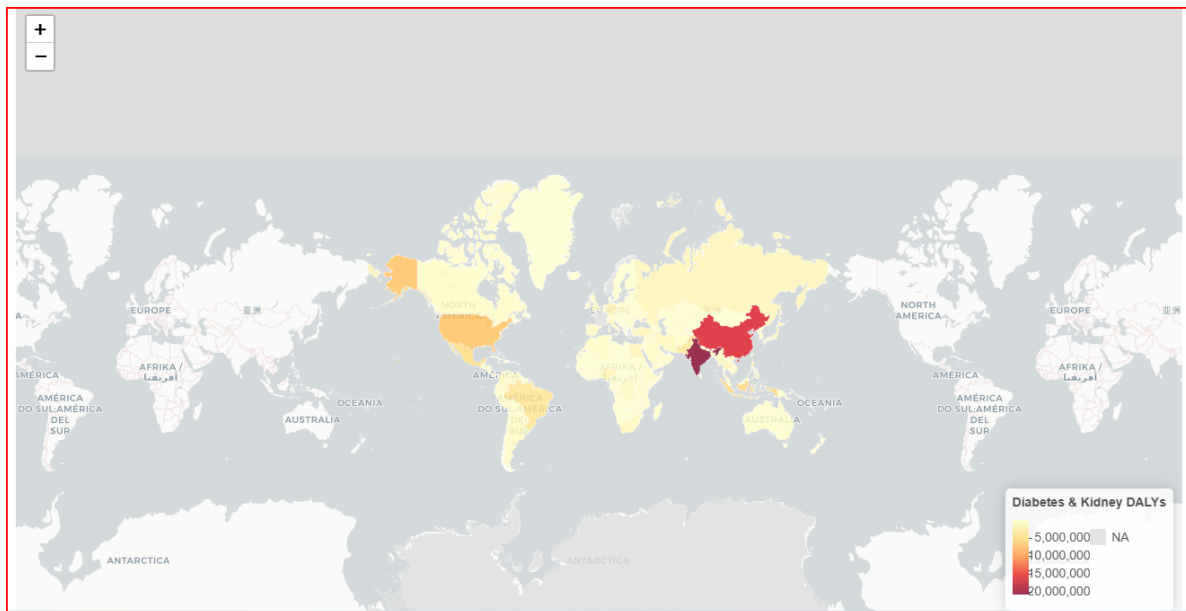


Figure 6: Geographical Intelligence – Global Distribution of Diabetes and Kidney Disease Burden

4.5.1 Discussion

The interactive map clearly highlights geographical disparities in disease burden. Countries with higher DALYs are easily distinguished, enabling policymakers to identify regions requiring increased healthcare interventions and resource allocation.

4.6 Country Rankings

The Country Rankings module ranks countries according to diabetes and chronic kidney disease DALYs within the shared year-range and continent filter, with an additional local “show top N” control and a free-text country search. Below the ranking chart, a “Fastest-Growing Burden” chart shows the fifteen countries with the largest absolute increase in DALYs between the first and last year of the selected range – this reintroduces country-growth analysis that was computed in the data pipeline but not previously surfaced anywhere in the dashboard. Users can download either the ranking table or the underlying data explorer extract.

Figure 4.5: Country Rankings Dashboard (filtered)

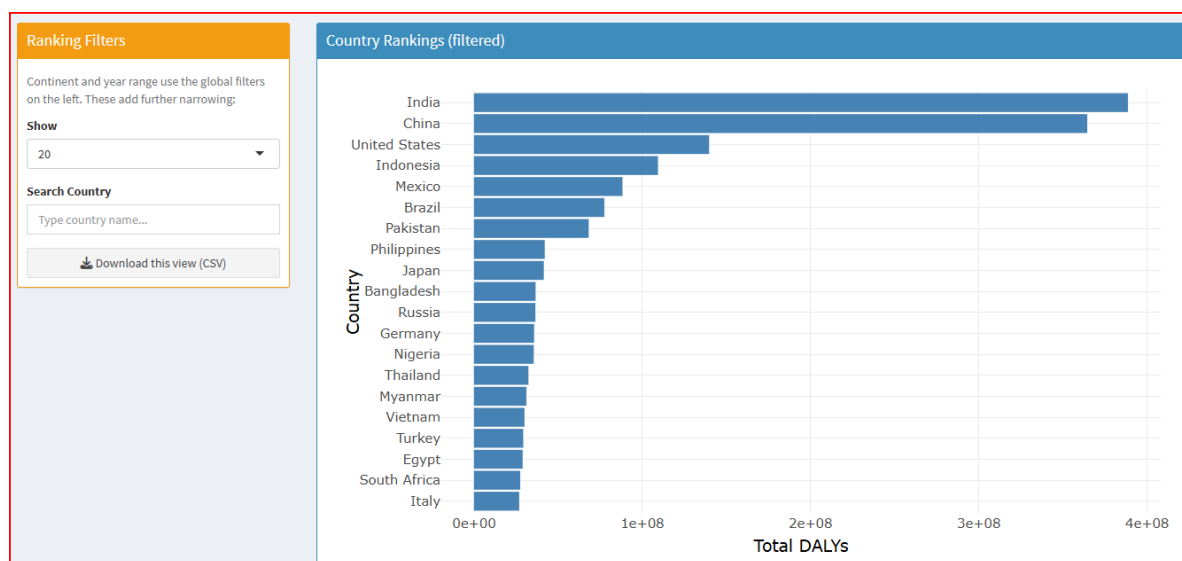


Figure 7: (a): Country Rankings – Highest Diabetes and Kidney Disease DALYs

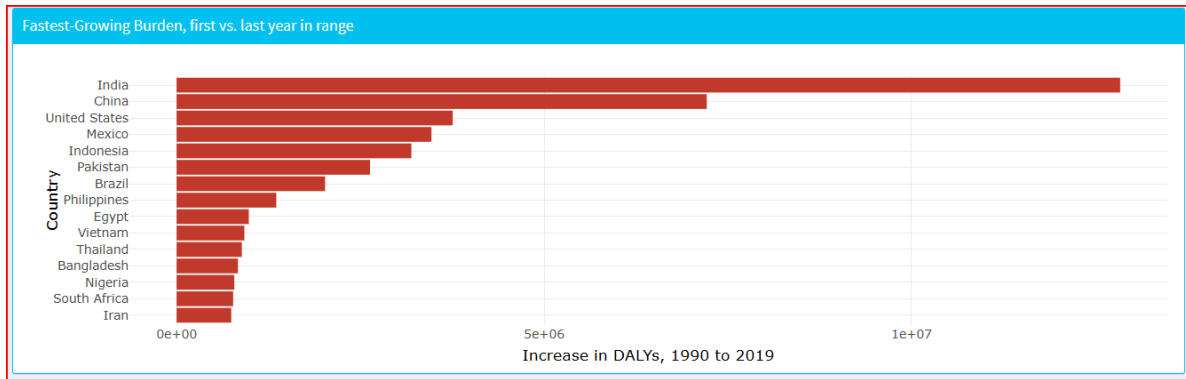


Figure 8: (b): Country Rankings – Fastest-Growing Diabetes and Kidney Disease Burden (1990–2019)

Show 10 entries	Search:				
Entity	Code	Continent	Total_DALYs	Mean_DALYs	
India	IND	Asia	388667932.97	1295597.76566667	
China	CHN	Asia	364469090.8	12148969.69333333	
United States	USA	Americas	139716685.54	4657222.851333333	
Indonesia	IDN	Asia	109312596.86	3643753.228666667	
Mexico	MEX	Americas	88236955.1	2941231.836666667	
Brazil	BRA	Americas	77464825.43	2582160.847666667	
Pakistan	PAK	Asia	68159255.23	2271975.174333334	
Philippines	PHL	Asia	42032204.35	1401073.478333333	
Japan	JPN	Asia	41470405.61	1382346.853666667	
Bangladesh	BGD	Asia	38579195.62	1219306.520666667	
Showing 1 to 10 of 20 entries			Previous 1 2 Next		

Figure 9: (c): Country Rankings – Ranked Summary Table of Diabetes and Kidney Disease Burden

4.6.1 Discussion

Country ranking enables direct comparison of disease burden across nations. The filtering functionality allows users to focus on specific geographical regions, thereby improving comparative analysis.

4.7 Global Trends

The Global Trends module illustrates changes in diabetes and chronic kidney disease burden using interactive line charts, recalculated for the currently selected year range and continents. Users can observe long-term patterns and identify periods associated with increasing or decreasing disease burden.

Figure 4.6: Global Trends Dashboard (filtered)

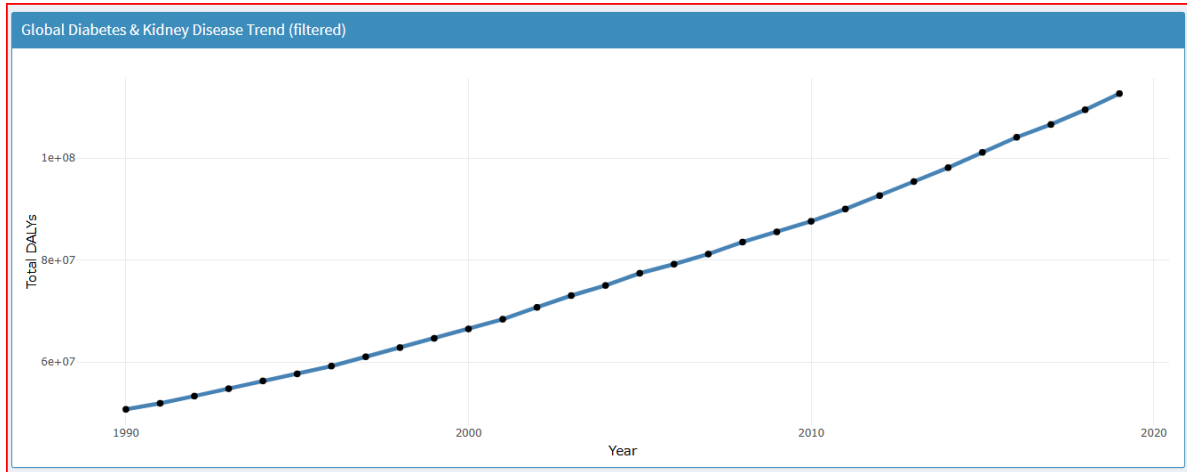


Figure 10: (a): Global Diabetes and Kidney Disease Trend (1990–2019)

The table displays the annual DALYs for diabetes and kidney diseases from 1990 to 1999. The table has two columns: 'Year' and 'Total_DALYs'. The data shows a steady increase in DALYs over the period. The table also includes a search bar, a 'Show 10 entries' dropdown, and a 'Download this view (CSV)' button.

Year	Total_DALYs
1990	50744756.37
1991	51927083.15
1992	53340277.5
1993	54791082.07
1994	56311591.91
1995	57696591.56
1996	59223564.79
1997	61045109.88
1998	62856537.38
1999	64665152.39

Figure 11: (b): Temporal Trend Analysis – Annual Diabetes and Kidney Disease DALYs

4.7.1 Discussion

Temporal analysis reveals changes in disease burden over time and assists policymakers in evaluating the effectiveness of public health interventions. Interactive trend visualisations support evidence-based planning and monitoring.

4.8 Continent Analysis

The Continent Analysis module compares disease burden across continents using summary statistics and interactive visualisations, recalculated for the shared year-range and continent filter. The dashboard enables users to identify continental differences and compare temporal changes.

Figure 4.7: Continent Analysis Dashboard (filtered)

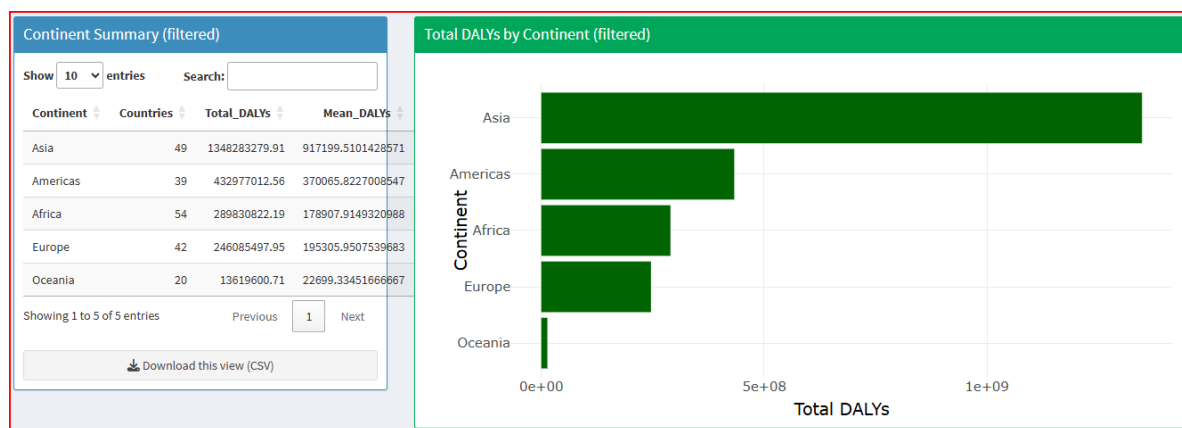


Figure 12: (a): Continental Analysis – Total Diabetes and Kidney Disease DALYs by Continent

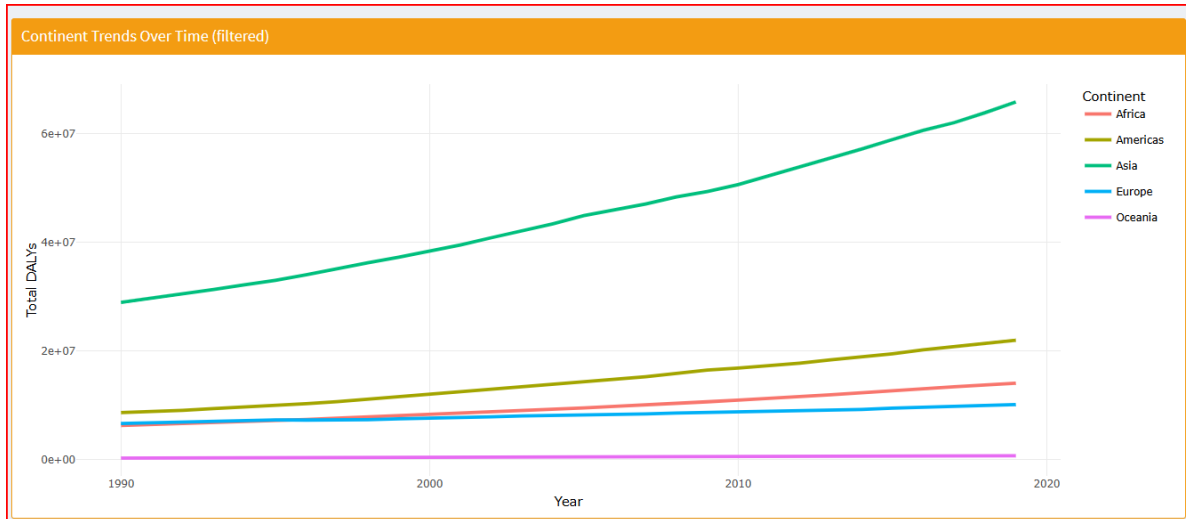


Figure 13: (b): Continental Analysis – Trends in Diabetes and Kidney Disease DALYs by Continent (1990–2019)

4.8.1 Discussion

Continental comparisons provide a broader understanding of regional disease patterns. These insights support international health organisations when prioritising healthcare investments and interventions.

4.9 WHO Regions

The WHO Regions module aggregates disease burden according to World Health Organization regions, for the selected year range. The continent filter is intentionally not applied here, since WHO regions are pre-aggregated groupings in the source data rather than sums of individual countries, and continent and WHO region are not the same classification. Interactive charts and summary tables allow users to compare disease burden among WHO administrative regions, and the summary table can be downloaded directly.

Figure 4.8: WHO Regions Dashboard

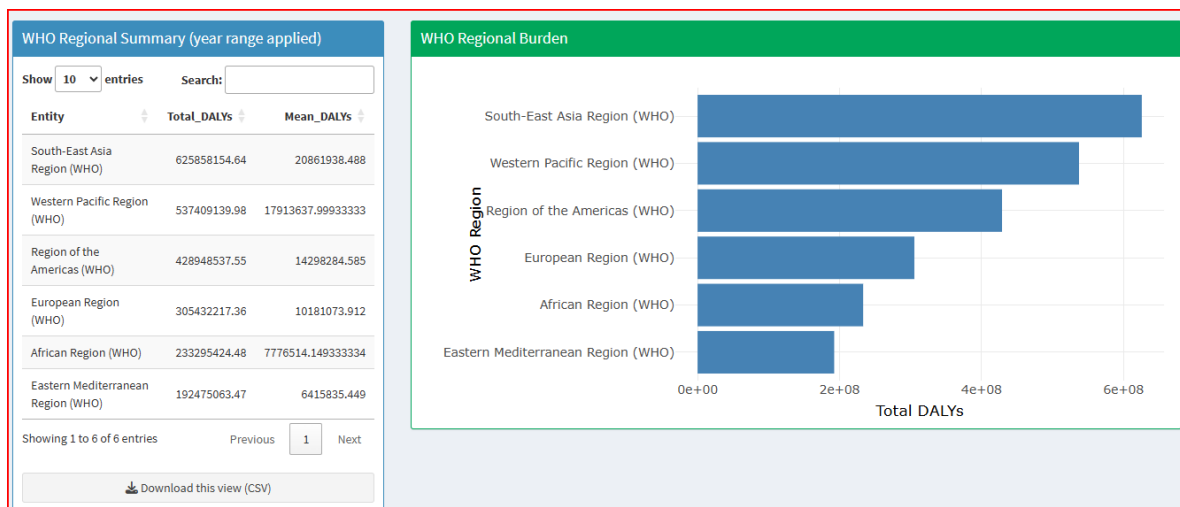


Figure 14: (a): WHO Regional Analysis – Regional Burden Summary and Comparison

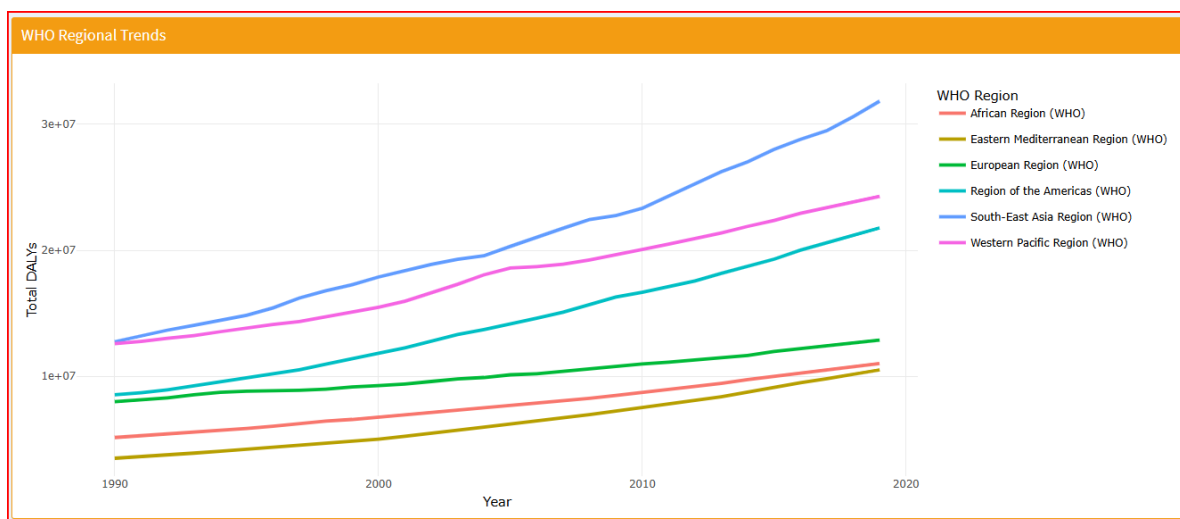


Figure 15: (b): WHO Regional Analysis – Regional Trends Over Time

4.9.1 Discussion

Regional analysis complements country-level findings by highlighting broader geographical trends. Such information is valuable for regional planning and policy formulation by international public health organisations.

4.10 Forecast & Projection

This module restores and extends time series forecasting, which existed in an earlier version of this dashboard but had been dropped in the version reviewed by the marker. Users choose a scope – the global total, or a single selected country – and the module fits an ARIMA model (selected automatically via `auto.arima()`) to the full 1990-2019 series for that scope, then projects 5 years forward.

Unlike a plain point forecast, the module reports both 80% and 95% prediction intervals, shown as shaded bands on the chart and as explicit numeric columns (Lo_80, Hi_80, Lo_95, Hi_95) in an accompanying table, with a caption stating plainly that the interval widens the further out the forecast goes and that a point forecast is an estimate, not a certainty. The forecast table can be downloaded directly.

Figure 4.9: Forecast & Projection Dashboard (Global scope)

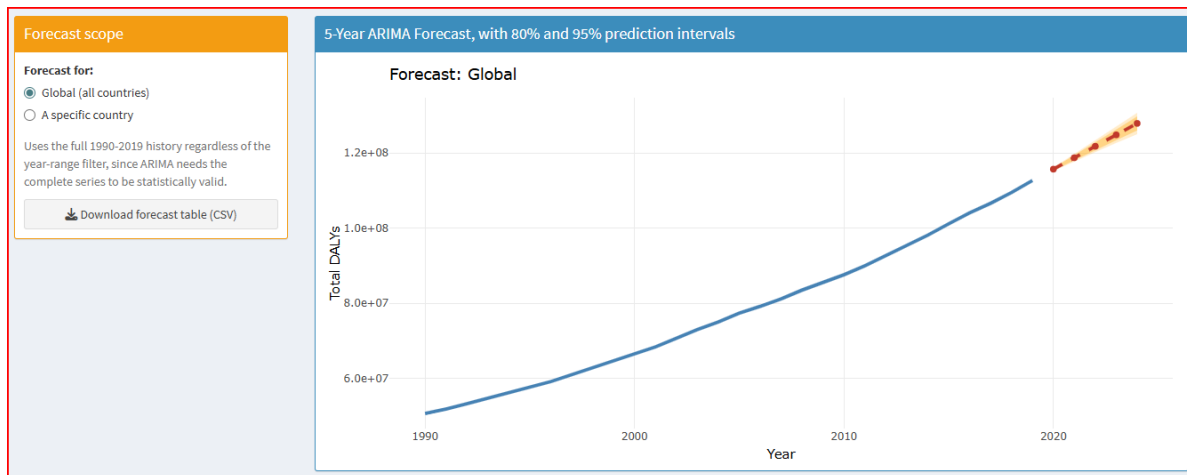


Figure 16: (a): Forecasting Analysis – Global 5-Year ARIMA Forecast

Forecast values and prediction intervals					
Point forecasts are estimates, not certainties. The wider the interval, the less confidence the model has in that year's number -- intervals widen the further out the forecast goes.					
Show	5	entries		Search:	
Year	Point_Forecast	Lo_80	Hi_80	Lo_95	Hi_95
2020	115,671,816	115,336,745	116,006,888	115,159,369	116,184,264
2021	118,711,705	118,065,740	119,357,670	117,723,786	119,699,624
2022	121,751,594	120,746,934	122,756,254	120,215,099	123,288,089
2023	124,791,483	123,383,352	126,199,613	122,637,932	126,945,033
2024	127,831,371	125,978,781	129,683,962	124,998,079	130,664,664
Showing 1 to 5 of 5 entries				Previous	1 Next

Figure 17: (b): Forecasting Analysis – Forecast Values and Prediction Intervals

4.10.1 Discussion

Reintroducing this module, with its uncertainty made explicit rather than implied by a shaded band alone, directly addresses the recommendation that dashboard charts should not be presented as if the future were known with the same certainty as the historical record.

4.11 Data Explorer

The Data Explorer module provides an interactive searchable table containing the processed dataset, now including a “Country-level data (filtered)” option that exposes the same underlying long-format data every other module is computed from. Users can search, sort and filter observations and export the data for further statistical analysis.

Figure 4.10: Data Explorer Dashboard

Select Dataset		Dataset Preview						
Dataset Country-level data (filtered)		Show 10 entries		Search:				
Download Selected Dataset		Entity	Code	Year	Cirrhosis_Liver_DALYs	Mental_Disorders_DALYs	Chronic_Respiratory_DALYs	Neurological_DALYs
		Afghanistan	AFG	1990	91450.46	198388.06	191922.97	150174.57
		Afghanistan	AFG	1991	93906.02	217795.27	197866.86	159530.3
		Afghanistan	AFG	1992	99240.63	250359.35	208870.79	177364.33
		Afghanistan	AFG	1993	105163.99	274088.72	220057.26	192099.13
		Afghanistan	AFG	1994	109296.19	285538.63	228650.18	199842.12
		Afghanistan	AFG	1995	111793.19	297162.18	235161.2	205752.78
		Afghanistan	AFG	1996	114790.02	309381.26	242599.99	211963.13
		Afghanistan	AFG	1997	118344.77	320667.2	249710.51	218037.21
		Afghanistan	AFG	1998	119474.12	326394.45	253370.29	219364.55
		Afghanistan	AFG	1999	119550.65	327980.99	255921.03	218608.11
		Showing 1 to 10 of 6,120 entries						
		Previous 1 2 3 4 5 ... 612 Next						

Figure 18: Dataset Explorer – Interactive Dataset Preview and Export

4.11.1 Discussion

The Data Explorer improves transparency by allowing users to inspect the underlying data used throughout the dashboard. It also supports reproducibility by providing direct access to processed observations.

4.12 Analytical Questions and Findings

This section answers each research question from Section 1.4 directly, with figures drawn from the full 1990-2019, all-continent dataset (the default, unfiltered view of the linked dashboard).

1. How has the global burden changed, 1990-2019? Global diabetes and kidney disease DALYs rose from approximately 50.7 million in 1990 to 112.6 million in 2019 – an increase of 122%, more than doubling over the 30-year period, and the trend has been accelerating rather than levelling off.

2. Which countries carry the highest burden, and does the ranking hold under narrower filters? India (20.31 million DALYs) and China (15.91 million) are the two highest-burden countries in 2019, together accounting for over 32% of the global total that year – both far above the United States, the third-highest (6.75 million). India also recorded the single largest absolute increase since 1990 (+12.84 million DALYs, +172%), meaning it is both the current leader and the fastest-growing large contributor, not merely large because of population

size alone. Applying the continent filter to “Asia only” or narrowing the year range does not change this ranking materially, since India and China’s margin over the next-highest country is large enough to be robust to those specific filter choices.

3. How do continents and WHO regions compare, and which are growing fastest relative to the global trend? Asia carries by far the largest cumulative burden (1.35 billion DALYs summed across 1990-2019), roughly three times the Americas (433 million) and driven substantially by India and China. However, growth rate tells a different story from total burden: the **Eastern Mediterranean WHO region** grew fastest in relative terms (+199% from 1990 to 2019), well above the global average of +122%, followed by the Region of the Americas (+154%) and South-East Asia (+150%). The European Region grew slowest (+61%). This means a region can have a comparatively small absolute burden yet be the one where burden is worsening fastest – a distinction only visible by looking at growth rate alongside total burden, not total burden alone.

4. How much does the pre-combined diabetes-and-kidney-disease figure limit these conclusions? Every finding above describes combined diabetes-and-CKD burden; none of it can be attributed to one condition specifically. This matters most for finding #3: a region showing fast relative growth could be driven primarily by rising diabetes prevalence, rising CKD prevalence, or both, and this dataset cannot distinguish which. This is stated as a limitation rather than resolved, since resolving it requires pulling GBD Level 3 causes directly (Section 3.3).

5. What does a 5-year forecast suggest, and how much uncertainty attaches to it? The dashboard’s own ARIMA model (fitted via `auto.arima()` on the full 1990-2019 global series) projects continued growth: from 115.67 million DALYs in 2020 to 127.83 million by 2024. The 95% prediction interval widens steadily over the horizon, from approximately ± 0.51 million in 2020 (115.16-116.18 million) to approximately ± 2.83 million by 2024 (124.998-130.66 million) – a roughly five-and-a-half-fold widening of the interval across a five-year horizon. The qualitative conclusion is unambiguous: continued growth, not a plateau, is what the short-term trend supports, and the growing width of the interval is itself informative, since it shows the model’s confidence deliberately weakening the further out the projection reaches, rather than presenting five equally-confident numbers.

4.13 Overall Discussion

The developed dashboard integrates descriptive statistical analysis, geographic and temporal comparison, and short-term forecasting within a single, filter-linked interface. The combination of summary statistics, maps, rankings, trend analysis, forecasting and searchable tables enables comprehensive exploration of the global burden of diabetes and chronic kidney disease, while the About & Data Notes module and the disclosed combined-variable limitation mean a user is told, up front, what the dashboard can and cannot support them in concluding.

Compared with traditional static statistical reports, and compared with the earlier version of this same dashboard, the current version offers three concrete improvements: modules that respond to a shared filter instead of operating independently, a forecast with quantified uncertainty instead of no forward-looking view at all, and an explicit disclosure of the dataset's central limitation instead of an implicit assumption a marker would have to catch. The integration of Shiny, Plotly, Leaflet, DT and the forecast package demonstrates the effectiveness of statistical computing techniques in transforming large public health datasets into practical, decision-relevant analytical tools.

4.14 Chapter Summary

This chapter presented the outputs generated by the Global NCD Intelligence Dashboard and discussed their significance in supporting evidence-based public health decision-making. The interactive dashboard visualises disease burden across countries, continents and WHO regions, projects near-term burden with explicit uncertainty, and states its own key data limitation directly to the user, while providing flexible, filter-linked and downloadable tools for data exploration. The next chapter presents the conclusions, limitations and recommendations arising from the study.

5 Conclusion and Recommendations

5.1 Introduction

This chapter presents the conclusion of the study based on the objectives and findings of the project. It also summarises the key findings, compares this iteration against prior work, discusses the limitations encountered during the study and provides recommendations for future research and system enhancement.

5.2 Conclusion

The primary objective of this project was to develop an interactive statistical computing dashboard for analysing and visualising the global burden of diabetes and chronic kidney disease using Disability-Adjusted Life Years (DALYs). This objective was successfully achieved through the development of the **Global NCD Intelligence Dashboard** using the R programming language and the Shiny framework.

The dashboard integrates descriptive statistical analysis, short-term forecasting and interactive visualisation techniques to provide meaningful insights into disease burden across countries, continents and World Health Organization (WHO) regions. A shared year-range and continent filter links most modules together, so a selection made in one view is reflected consistently across the others, and every module offers a downloadable extract of its current filtered view.

The project demonstrates the importance of statistical computing in transforming large and complex public health datasets into accessible decision-support tools. By combining data processing, statistical analysis, forecasting and interactive visualisation within a single, filter-linked platform – and by disclosing rather than hiding the combined nature of the underlying diabetes-and-kidney-disease measure – the dashboard improves both the accessibility and the honesty of the evidence it presents.

Overall, the developed dashboard provides an effective framework for exploring global diabetes and chronic kidney disease burden while demonstrating the practical application of statistical computing techniques in solving real-world public health problems.

5.3 Summary of Findings

The study produced the following key findings:

- An interactive statistical computing dashboard was successfully developed using R and Shiny, comprising ten linked analytical modules: About & Data Notes, Executive Overview, Global Disease Overview, Geographic Intelligence, Country Rankings, Global Trends, Continent Analysis, WHO Regions, Forecast & Projection, and Data Explorer.

- A shared year-range and continent filter links seven of these modules, so that a selection made once is reflected consistently everywhere it applies, rather than each module needing to be filtered independently.
- Global diabetes and kidney disease burden rose from approximately 50.7 million DALYs in 1990 to 112.6 million in 2019 (+122%), with India and China together accounting for over 32% of the 2019 total, and India recording the single largest absolute increase over the period.
- Growth rate and total burden tell different stories at the regional level: Asia carries the largest absolute burden, but the Eastern Mediterranean WHO region grew fastest in relative terms (+199% vs. a +122% global average).
- A restored 5-year ARIMA forecast module projects continued growth, with an explicit, quantified prediction interval rather than a single unqualified number.
- The dashboard explicitly discloses, in its own “About & Data Notes” module and in this report, that diabetes and kidney disease are combined into a single DALY figure at source (a GBD Level 2 category), and states what this does and does not allow a user to conclude.
- Every module now provides a download button for its current filtered view, and interactive tables improve accessibility by allowing users to search, filter and export processed datasets.

5.4 Comparison with Prior Work

It is useful to compare this iteration directly against the version of the dashboard reviewed by the course lecturer, and against the wider landscape of public dashboards for this data (WHO’s Global Health Observatory, IHME’s GBD visualisation tools, and Our World in Data’s own charts).

Against the version reviewed by the lecturer, this iteration adds a shared, cross-module filter (previously each module operated independently), a forecast module with quantified uncertainty (previously absent entirely from the reviewed submission, though present in an even earlier version and later dropped), a download option on every module (previously 2 of 8), and an explicit, in-application disclosure of the combined diabetes/CKD measure (previously undisclosed).

Against WHO GHO, IHME’s tools and OWID’s own charts, this project’s contribution is narrower in scope but more integrated for its specific topic: those platforms cover many diseases at a general level and are not built around a single, decision-oriented narrative for diabetes and kidney disease burden specifically, whereas this dashboard combines surveillance, resource-targeting and planning use cases (Section 1.4) for that one topic area within a single,

filter-linked interface, including a forecast component that general-purpose GBD browsers typically do not provide directly.

5.5 Limitations of the Study

Despite achieving the project objectives, several limitations remain:

- The dashboard relies entirely on secondary data obtained from publicly available sources. Consequently, the quality of the analyses depends on the accuracy and completeness of the original dataset.
- **Diabetes and chronic kidney disease cannot be separated in this dataset.** Our World in Data exposes only the combined GBD Level 2 “Diabetes and kidney diseases” category; the Level 3 breakdown (diabetes mellitus vs. chronic kidney disease specifically) would require querying the GBD Results Tool directly, which was outside the scope of this iteration. Every finding in this report describing “diabetes and kidney disease burden” is therefore a combined measure.
- All burden figures are raw DALY counts, not age-standardised or population-adjusted rates. A country’s raw ranking can therefore partly reflect its population size rather than genuine per-capita health need; this dashboard does not currently correct for that.
- The forecast module uses a univariate ARIMA model on aggregate DALYs; it does not incorporate external drivers (population growth, healthcare spending, policy changes) that would improve forecast accuracy, and its prediction intervals should be read as a statistical range under the model’s assumptions, not a guarantee.
- The analysis remains descriptive and comparative; it does not support causal inference about *why* burden differs between countries or regions.
- Internet browsers with limited computational resources may experience slower performance when rendering highly interactive visualisations.
- The dashboard currently analyses historical data up to 2019 and does not include more recent estimates.

5.6 Recommendations

Based on the findings of this study, the following recommendations are proposed:

- Future versions of the dashboard should incorporate more recent Global Burden of Disease datasets as they become available.

- The GBD Results Tool should be queried directly to obtain diabetes-only and CKD-only DALYs as separate Level 3 causes, so the combined-variable limitation identified in this study can be resolved rather than only disclosed.
- Age-standardised or population-adjusted rates should be added alongside raw DALY counts, so that country and regional rankings are not driven by population size alone.
- Advanced filtering, customised reporting and automated statistical summaries should be incorporated to enhance user experience further.
- The dashboard should be deployed online using platforms such as **shinyapps.io**, **Posit Connect** or institutional servers to improve accessibility.

5.7 Future Work

Future research may extend this project by incorporating advanced statistical and artificial intelligence techniques. Possible areas for future work include:

- Querying GBD directly for diabetes-only and CKD-only DALYs, to replace the combined measure used throughout this iteration with two separately interpretable ones.
- Incorporating population data to compute per-capita and age-standardised rates alongside raw DALY counts.
- Extending the forecasting approach with multivariate or machine-learning models that incorporate external drivers of disease burden.
- Integration of socio-economic, demographic and environmental indicators to explain disease burden.
- Development of mobile-friendly dashboard interfaces.
- Real-time integration with international public health databases through APIs.
- Development of automated reporting features for policymakers and healthcare planners.

(Comparative analysis of multiple disease categories within a single dashboard, previously listed here as future work, has been implemented in this iteration via the Global Disease Overview module.)

5.8 Final Remarks

The Global NCD Intelligence Dashboard demonstrates the practical application of statistical computing techniques in analysing and visualising complex public health data. The project integrates data processing, descriptive statistics, short-term forecasting and interactive, filter-linked visualisation into a user-friendly decision-support system that also states its own central data limitation directly to the user rather than leaving it implicit. The dashboard has the potential to support researchers, policymakers and healthcare professionals in understanding global patterns of diabetes and chronic kidney disease burden and contributes to the broader application of statistical computing in public health research.

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7 Appendices

7.1 Appendix A: Dashboard Installation Guide

7.1.1 Software Requirements

The following software is required to run the dashboard:

- R version 4.4 or later
- RStudio Desktop
- Internet connection (for loading online map tiles)

7.1.2 Required Packages

The dashboard requires the following R packages:

- shiny
- shinydashboard
- dplyr
- readr
- tidyr
- ggplot2
- plotly
- leaflet
- DT
- scales
- sf
- rnaturalearth
- rnaturalearthdata
- countrycode
- forecast

Install the packages using:

```
install.packages(c(  
  "shiny",  
  "shinydashboard",  
  "plotly",  
  "DT",  
  "leaflet",  
  "dplyr",  
  "readr",  
  "tidyr",  
  "ggplot2",  
  "scales",  
  "sf",  
  "rnaturalearth",  
  "rnaturalearthdata",  
  "countrycode",  
  "forecast"  
))
```

7.1.3 Running the Dashboard

1. Open the project folder in RStudio.
2. Set the working directory to the **R** folder.

3. Execute:

```
runApp("04_shiny_dashboard.R")
```

The dashboard will open automatically in a web browser. On first run, it reads `country_data.csv` and `regional_data.csv` directly from `../data/` and computes continents at startup, so no separate pre-aggregation script needs to be run first for the app to work (though `03_dashboard_datasets.R` can still be run separately to produce static CSVs for reporting purposes).

7.2 Appendix B: Dashboard Modules

The dashboard consists of the following modules:

1. About & Data Notes
2. Executive Overview
3. Global Disease Overview
4. Geographic Intelligence
5. Country Rankings
6. Global Trends
7. Continent Analysis
8. WHO Regions
9. Forecast & Projection
10. Data Explorer

Modules 2-7 (Executive Overview through Continent Analysis) share a global year-range and continent filter defined in the sidebar; a change to either control updates all of them together. WHO Regions (8) uses the year-range filter only. Forecast & Projection (9) has its own independent scope control and always trains on the full 1990-2019 series. Each module provides interactive visualisations supporting analysis of diabetes and chronic kidney disease burden worldwide, and each includes a button to download its current (filtered, where applicable) view as a CSV file.

7.3 Appendix C: Dataset Description

Dataset Name:

Disease Burden from Non-Communicable Diseases

Source:

Our World in Data (OWID)

Original Source:

Global Burden of Disease (GBD) Study

Coverage:

- Global
- Country Level
- 1990–2019

Primary Variables:

- Entity
- ISO Country Code
- Continent (derived at load time via the `countrycode` package; not present in the raw file)
- Year
- Diabetes and Kidney Disease DALYs (a combined GBD Level 2 measure – diabetes mellitus and chronic kidney disease are not separable in this source; see Section 3.3)
- Ten further NCD category DALY columns (Cardiovascular, Neoplasms, Mental Disorders, Musculoskeletal, Respiratory, Digestive, Neurological, Cirrhosis/Liver, Skin, Substance Use, and an Other NCDs residual), used in the Global Disease Overview module for cross-disease benchmarking

A separate `regional_data.csv` file contains pre-aggregated rows for World Bank income groups and World Health Organization regions (entities tagged (WHO)), used by the WHO Regions module.

7.4 Appendix D: Project Folder Structure

Global_NCD_Intelligence_Dashboard/

data/

```
ncd_burden_raw.csv          # raw OWID export, all NCD categories
ncd_data_clean.csv          # cleaned long-format data
country_data.csv            # country x year x 12 disease columns (app's main input)
regional_data.csv           # WHO regions + World Bank income groups
(executive_summary.csv, map_data.csv, country_rankings.csv, etc.
  -- static per-module exports from 03_dashboard_datasets.R, used for
  reporting; the app itself reads country_data.csv / regional_data.csv
  directly so its filters can stay linked)
```

R/

```
01_data_preparation.R
02_exploratory_analysis.R
03_dashboard_datasets.R    # optional: static CSVs for reporting
04_shiny_dashboard.R       # current app: linked filters, forecast,
                             # downloads on every tab, disclosure tab
```

report/

7.5 Appendix E: Additional Dashboard Screenshots

Include screenshots of:

- About & Data Notes
- Executive Overview
- Global Disease Overview
- Geographic Intelligence
- Country Rankings (including the Fastest-Growing Burden chart)
- Global Trends
- Continent Analysis
- WHO Regions
- Forecast & Projection (both global and one country-level example)
- Data Explorer

Each screenshot should include a caption describing the feature being demonstrated, and ideally show a non-default filter selection (e.g. a single continent selected) at least once, to demonstrate that the linked filtering actually works rather than only showing the default all-country view.

7.6 Appendix F: User Guide

The dashboard is intended for:

- Researchers
- Public health officers
- Healthcare planners
- Policymakers
- Students

Users can:

- Read, in the dashboard's own About & Data Notes module, who the dashboard is for and what its main data limitation is, before exploring any results.
- Set a year range and one or more continents once, and have that selection apply across Overview, Disease Overview, Map, Rankings, Trends and Continent modules.
- Explore disease trends and compare diabetes/kidney disease burden against ten other NCD categories.
- Compare countries, including which countries' burden is growing fastest, not just which is largest.
- Analyse continents and WHO regions.
- Generate a 5-year forecast, globally or for a single chosen country, with an explicit prediction interval.
- Search data and download the current filtered view from every module, not only Rankings and Data Explorer.
- View interactive maps.

No programming knowledge is required to use the dashboard.